

LAB 9

Introduction to Object-Oriented Programming

{OOP}

Thread

Lab09 Exercises

❖ Lab09.1. Multiple Processes running parallelly

Write a program to create a class to handle multiple task processes at the same time by using Thread.

- Use **Thread class** or **Runnable interface** to create sub-process
- Use **Thread.sleep()** to create fake task delay duration
- Refresh the percentage of task completion in every second

- Create a “Task” class with following abstract class

```
abstract class SubTask {  
    int duration;  
    String name;  
}
```

- Create a thread class to handle tasks

```
class TaskProcess implements Runnable {  
    Task task;  
    TaskProcess(Task task) {  
        this.task = task;  
    }  
  
    @Override  
    public void run() {  
        // to do  
    }  
}
```

Console:

```
::: Task Handler :::
```

```
=====
| No          | Name                | Duration  |
=====
| 1           | Format hard disk    | 1000 mills|
| 2           | Download video      | 6000 mills|
| 3           | Develop an app      | 8000 mills|
=====
```

```
1. Add new task
2. Delete a task
3. Proceed tasks
Choose an opt: 3
```

Select tasks to perform parallel process:

```
Input task No: 1
Do you want to add more task (y/n)?: y
Input task No: 2
Do you want to add more task (y/n)?: y
Input task No: 3
Do you want to add more task (y/n)?: n
```

```
1s :: Develop an app - 12.50% - processing ::
1s :: Format hard disk - 100.00% - completed ::
1s :: Download a video - 16.67% - processing ::
2s :: Download a video - 33.33% - processing ::
2s :: Develop an app - 25.00% - processing ::
3s :: Download a video - 50.00% - processing ::
3s :: Develop an app - 37.50% - processing ::
4s :: Download a video - 66.67% - processing ::
4s :: Develop an app - 50.00% - processing ::
5s :: Download a video - 83.33% - processing ::
5s :: Develop an app - 62.50% - processing ::
6s :: Download a video - 100.00% - completed ::
6s :: Develop an app - 75.00% - processing ::
7s :: Develop an app - 87.50% - processing ::
8s :: Develop an app - 100.00% - completed ::
```

Good luck 🍀

👉 How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don't zip the files for submission